

CS3240

AI in Everyday APPS

This work is part of an in-progress research work.
It is shared here FOR its academic relevance to the course CS3240.
DO NOT DISTRIBUTE or FORWARD it in any form without taking an explicit consent.

S2AY2025-26

Assoc Prof Bimlesh Wadhwa
bimlesh@nus.edu.sg

Materials used for this course are to be strictly used only for the purpose of this course. These are NOT to be distributed or forwarded to anyone or posted on social media without consent.

Example

The Secret Sauce Behind Search Personalisation

GOJEK Engineering Blog, 2019

Link:

<https://www.gojek.io/blog/the-secret-sauce-behind-search-personalisation>

Gojek is a leading Southeast Asian super-app founded in Jakarta, Indonesia. It brings multiple everyday services into a single platform. Users can book rides, order food and groceries, send parcels, pay bills, and make cashless transactions through its integrated wallet and payment ecosystem. GoFood is Gojek's food-delivery service, allowing users to order meals from local restaurants and street-food vendors directly to their door.



By Jewel James

Are you looking for food? Thinking about burgers and crispy chicken fritters with creamy sauces sitting on a warm, soft bun? At least, that's what I am thinking about. May be you're different. Vegan perhaps? Or craving noodles? Each of us turn into a different person when we are hungry, and this is the story about how GoFood — Gojek's food delivery service — helps you find the bite you crave.

| *At the core of the food ordering experience, sits search.*

From the source article

Foodie Mila

24 years old, Student

"I know all the best restaurants around!"



1 Number of orders with
burger palace

4 Number of orders in
fastfood cuisine

2 Number of orders in
western cuisine

1 Number of orders in
chicken & duck cuisine

Workaholic Husain

35 yrs old, Manager

"Deliver my food fast!"



2 Number of orders with
miemie king

2 Number of orders in
korean cuisine

3 Number of orders in
noodles cuisine

1 Number of orders in
chicken & duck cuisine

This image shows **two contrasting GoFood user personas** with different food preferences and ordering behaviors.

Foodie Mila :a 24-year-old student.

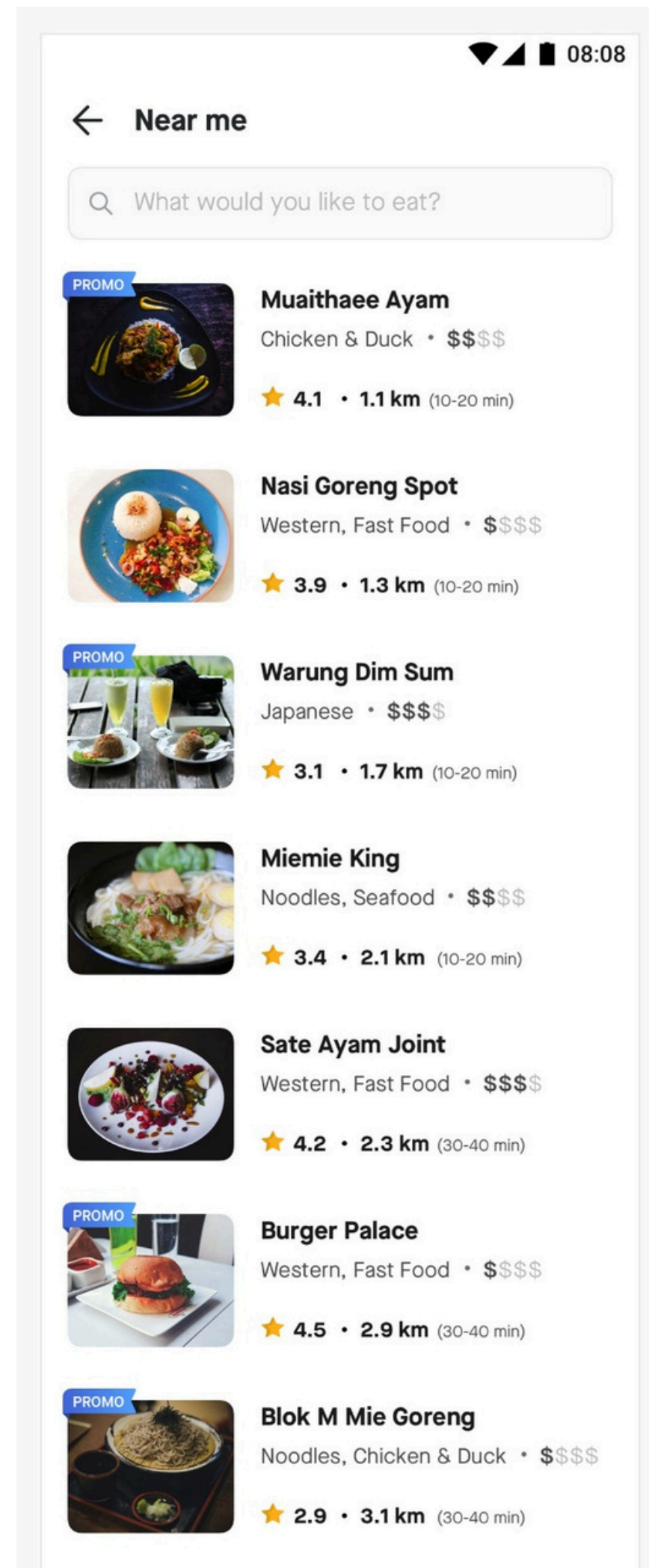
Her order history shows variety, with multiple orders across fast food, Western cuisine, and chicken and duck dishes.

Workaholic Husain: a 35-year-old manager.

His order history is more habitual, focusing on noodles, Korean food.

They both open our app and click on the “**NEAR ME**” title that lets users find the restaurants near them.

App shows them the nearest restaurants first, and this is what they both see.



Before personalization,

the “NEAR ME” feature used a **simple distance-based ranking**: it showed the closest restaurants first and then continued in an endless list as distance increased.

This design exposed users to **too many options with little guidance, forcing them to scroll and compare extensively**.

As a result, **the high cognitive load often led to decision fatigue and drop-off before placing an order**.



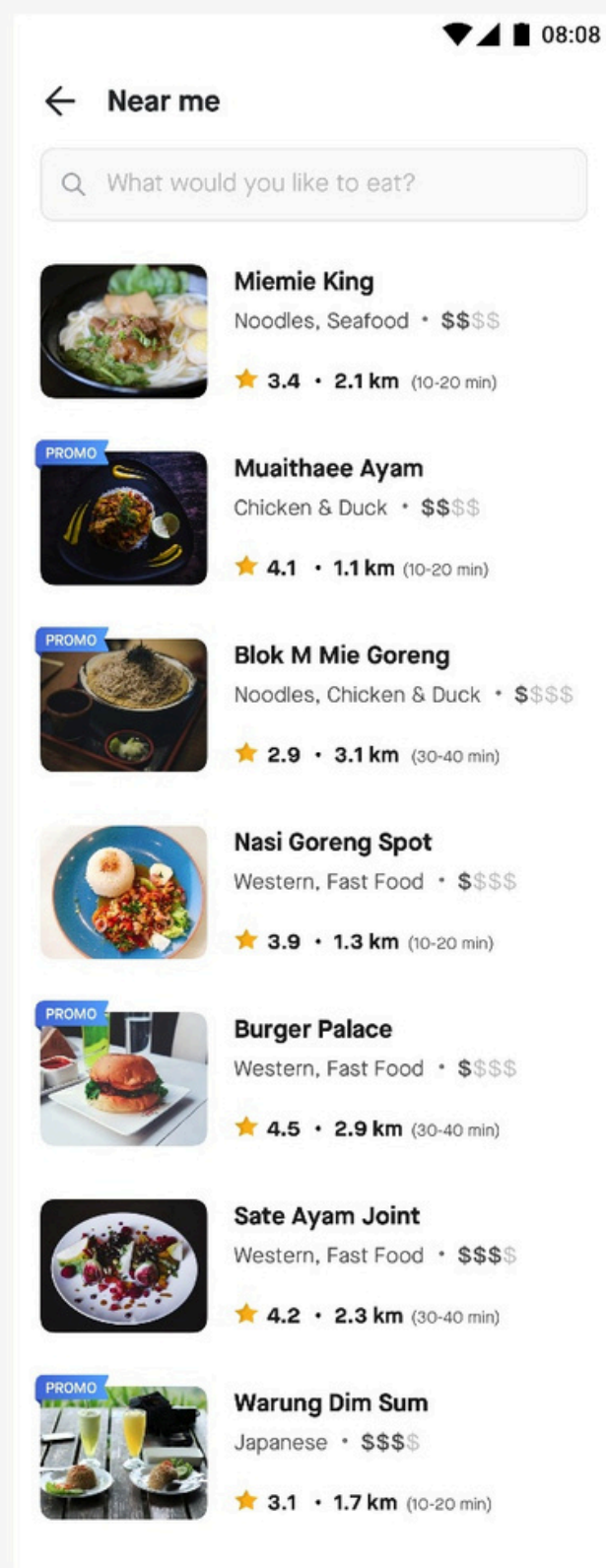
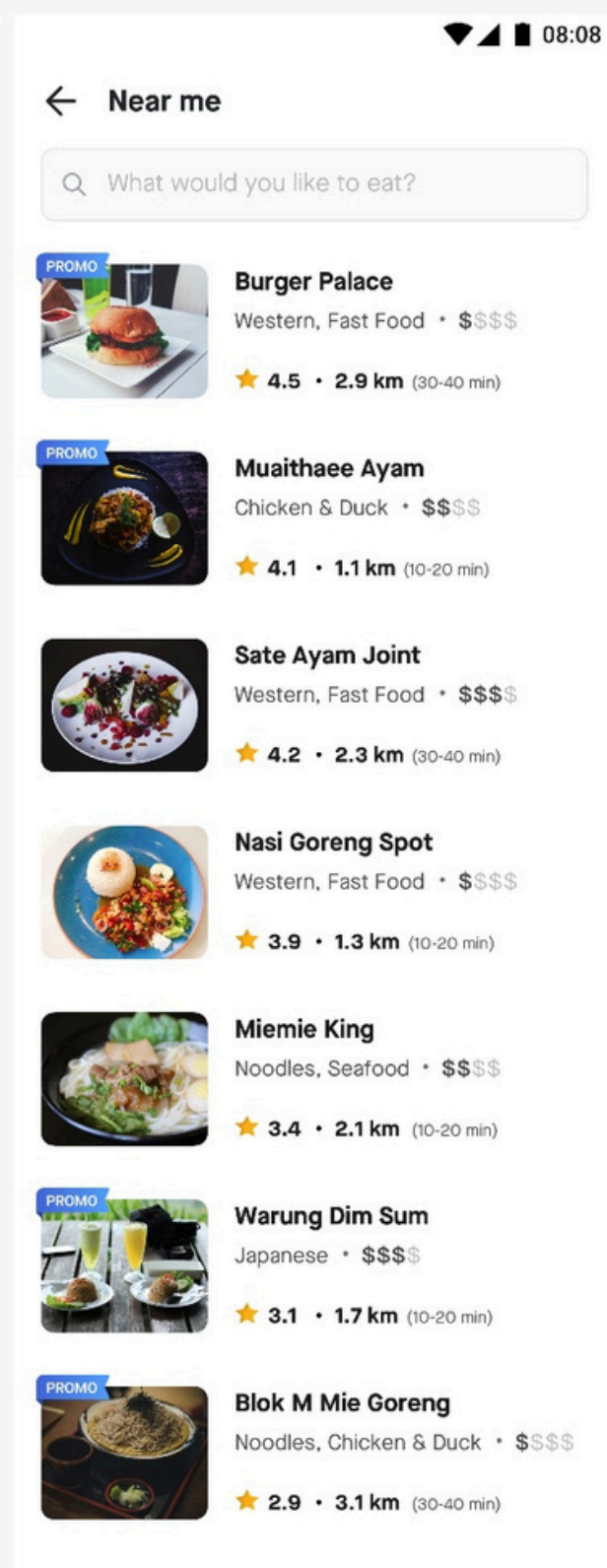
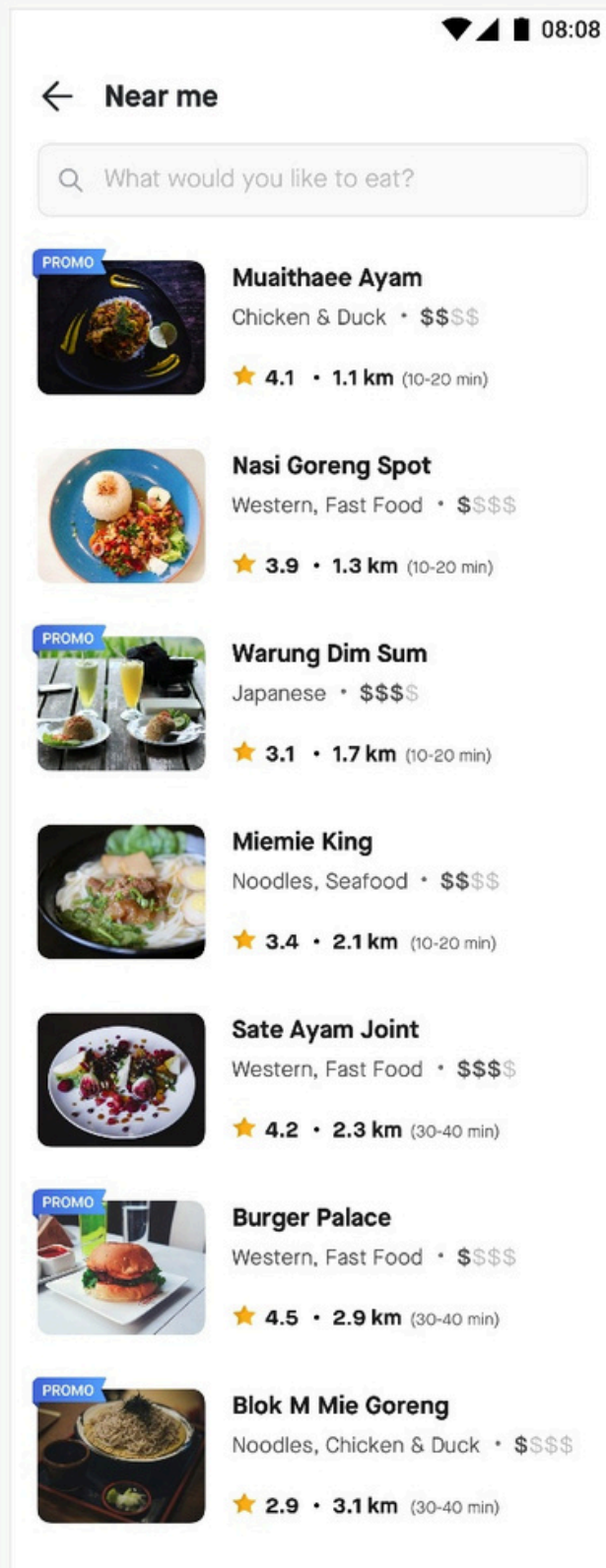
Before relevance ranking & personalisation
(For both)



After relevance ranking & personalisation
(For Mila)



After relevance ranking & personalisation
(For Husain)



After introducing **AI-driven relevance ranking** and **personalization**, the same entry point now produces different default results for different users.

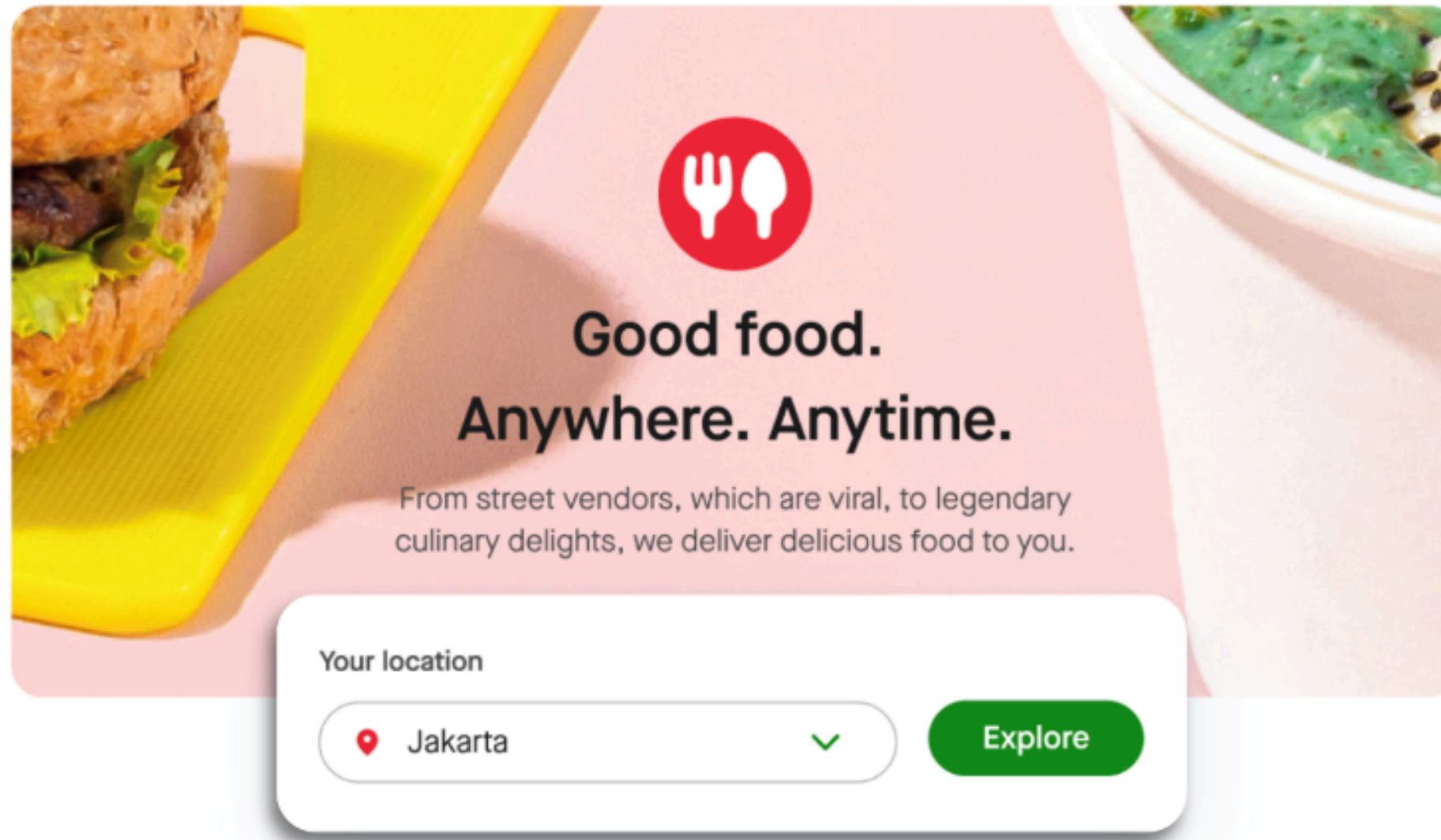
Instead of ranking restaurants purely by distance, **the system reorders the list based on each user's past behavior, cuisine preferences, and ordering patterns.**

For Mila, restaurants aligned with her interest in variety and fast food exploration appear higher by default, even if they are slightly farther away.

For Husain, the list prioritizes familiar noodle and quick-delivery options that match his habit of ordering fast and reliably.

This change transforms search into a **smart default** experience: **users no longer need to scroll extensively or compare many options.**

The **AI reduces cognitive load** by surfacing the most relevant choices first, **making decision-making faster while keeping the interface itself unchanged.**



Source: Google Images

This case is a strong example of **smart defaults** in action:

AI capabilities are surfaced through UI/UX by changing what users see by default, not by introducing new features or interfaces.

Effective AI in super-apps is often invisible improving decision-making by reducing scrolling, filtering, and cognitive load.

Key Insights:

Rather than adding explicit AI controls, intelligence is embedded into the default ordering of results, making the most relevant options appear first with no extra effort from the user.

AI is embedded in the default ranking of the “NEAR ME” list. Although different users tap the same entry point and see the same UI layout, the order of restaurants is personalized based on historical orders, cuisine preferences. This means **the interface looks the same, but what is shown first is different.**

Example

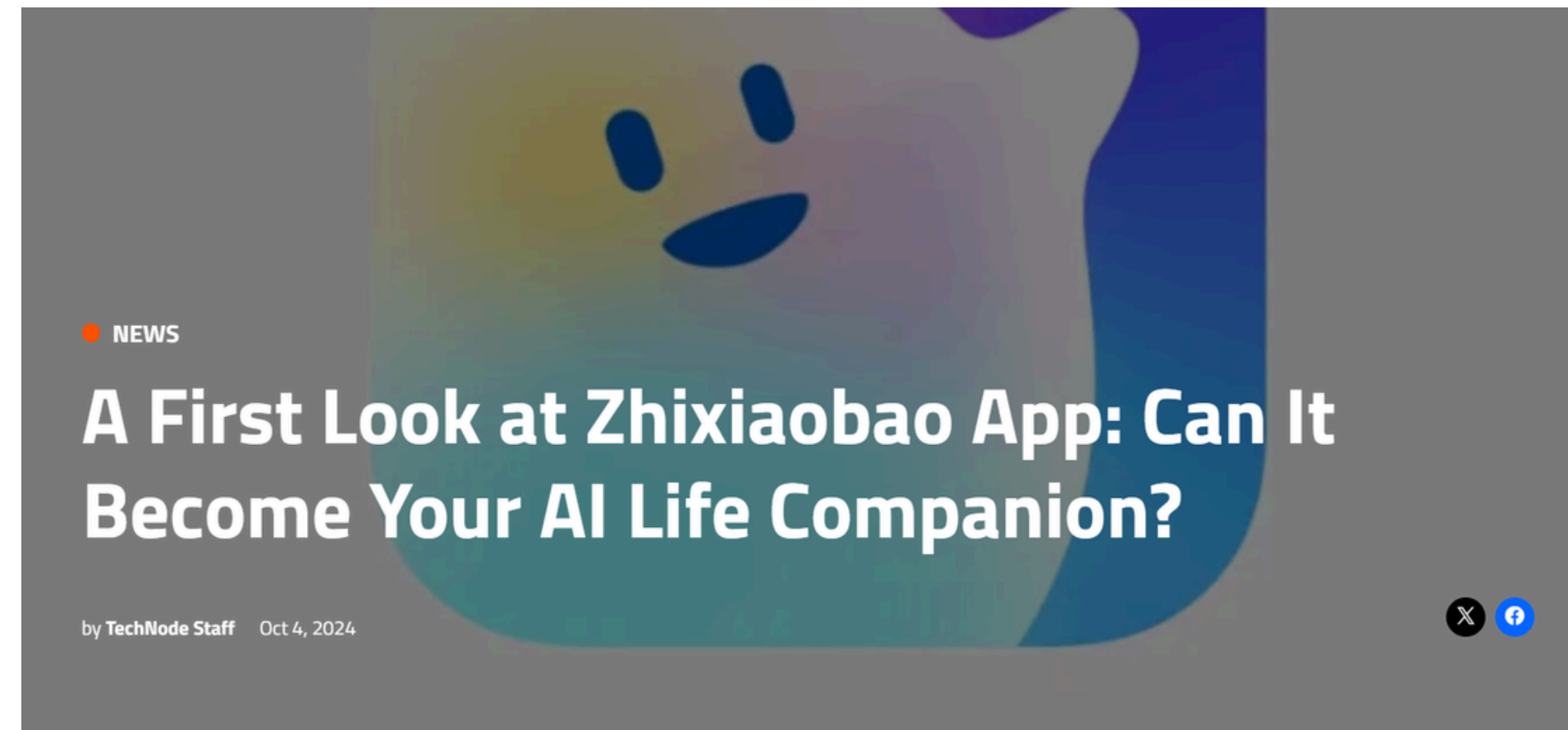
A First Look at Zhixiaobao App: Can It Become Your AI Life Companion?

GTechNode Staff, 2024

Link:

<https://technode.com/2024/10/04/a-first-look-at-zhixiaobao-app-can-it-become-your-ai-life-companion>

Zhixiaobao is an **AI assistant** developed by Alipay. It is available as a standalone app, and it can also be accessed directly inside Alipay by searching for its name. Within Alipay, Zhixiaobao functions as an AI interaction layer, helping users find and operate different services through conversation, cards, and smart actions.



A First Look at Zhixiaobao App: Can It Become Your AI Life Companion? Credit: Zhixiaobao

Note: The article was first published on [TechNode China](#) written by Steven Lee and translated by [Zinan Zhang](#).

At the 2024 INCLUSION in Shanghai, Ant Group unveiled its independent AI app, Zhixiaobao. Developed based on Ant's Bailing large language model, this new AI lifestyle assistant is integrated with Alipay, offering quick services like booking tickets, ordering food, accessing public transit codes, and finding nearby activities through conversational interactions.

LATEST STORIES

Meituan to acquire on-demand grocery firm Dingdong Maicai for \$717 million

Feb 5, 2026

Changan Automobile teams up with CATL to deploy sodium-ion battery cars in 2026

From the source article



This figure shows the main user interface of Zhixiaobao.

Left screen:

The entry screen welcomes users to Zhixiaobao. **A friendly AI character is used as a visual identity, and users can log in with their Alipay account.** This emphasizes that Zhixiaobao is closely integrated with the Alipay ecosystem.

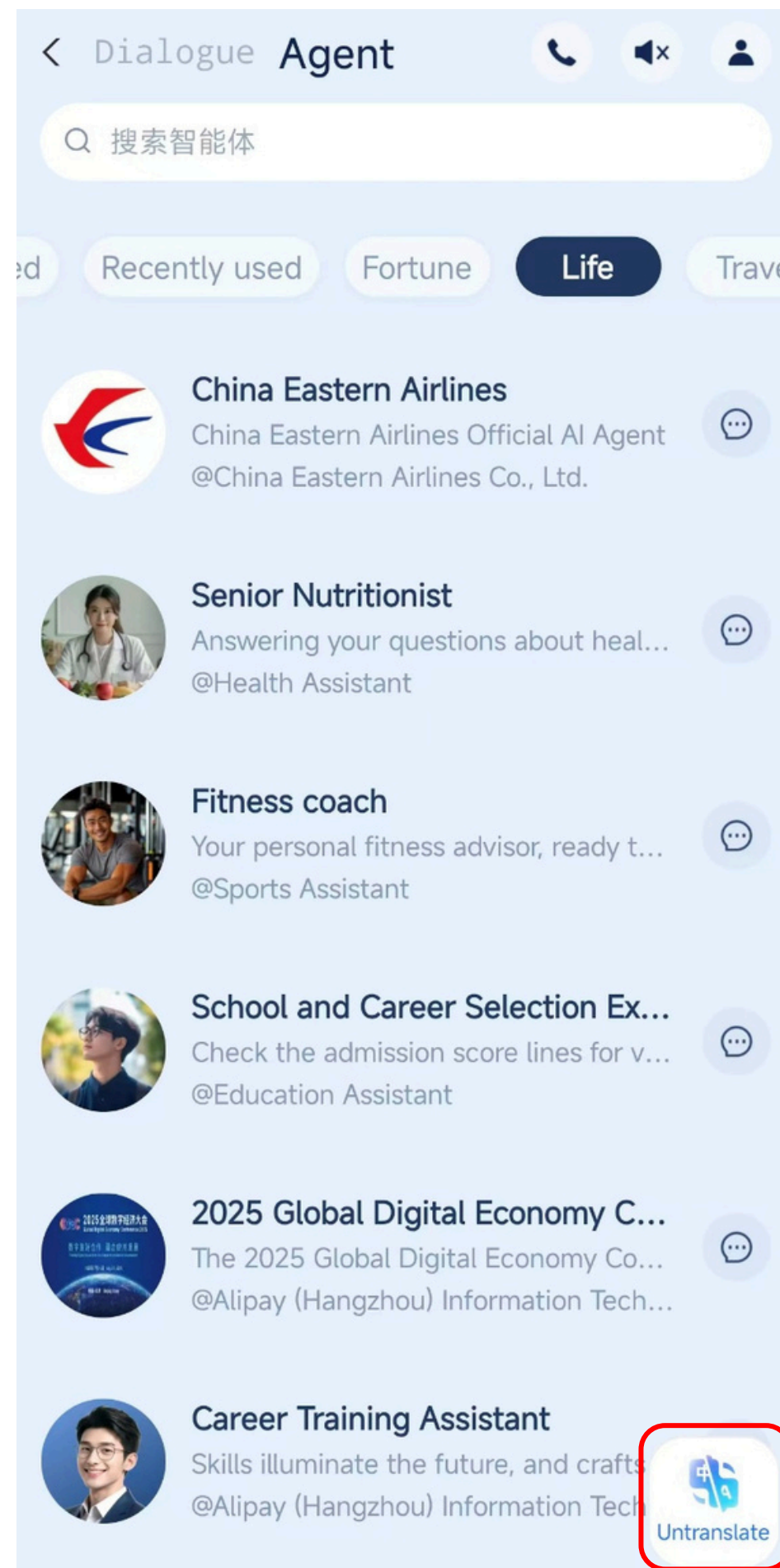
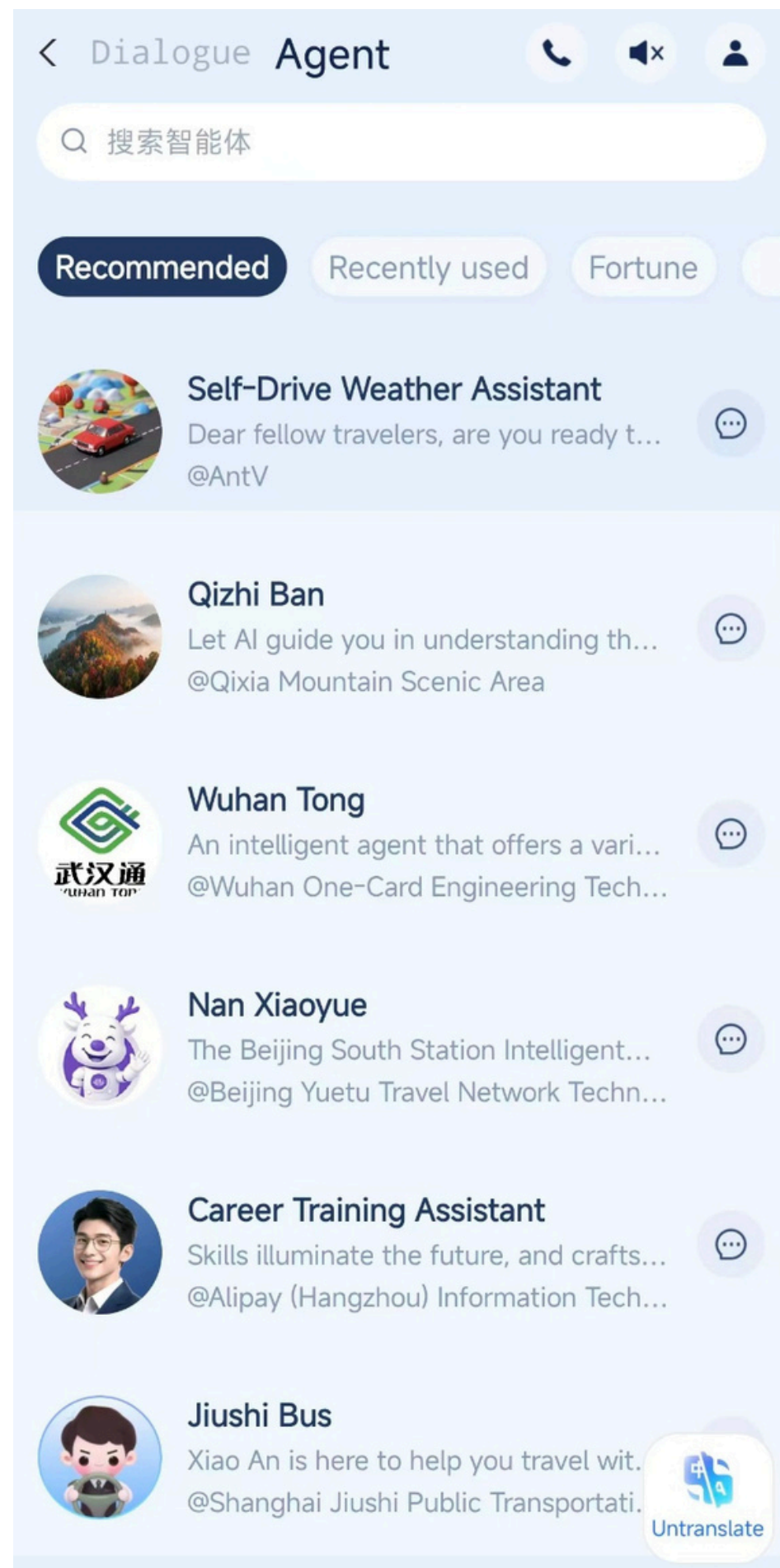
Middle screen:

The home view presents a set of AI-generated cards representing different services and tasks, such as transportation, delivery, or daily utilities. These cards act as clickable shortcuts, allowing users to jump directly to related services.

Right screen:

The assistant proactively provides context-aware alerts, such as a high-temperature weather warning and package delivery updates. Each alert is shown as a card with clear visuals and actionable information, demonstrating how the AI surfaces timely recommendations and reminders.

We couldn't find an English version of the image, so we've provided a description [here](#).



This screen shows the **“Intelligent Agents” list** in Zhixiaobao.

Different AI agents are grouped by topics such as fitness, travel, career, and daily advice.

Users can select an agent and start a conversation.

The real-time translation feature in Alipay is well designed. It exists as a small, draggable widget.

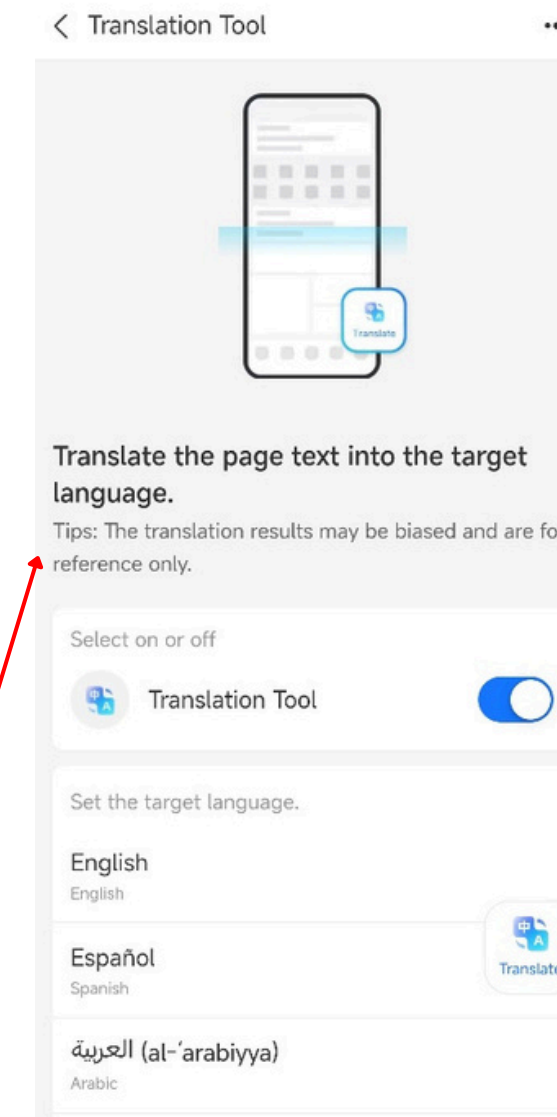
A single click switches between Chinese and English, and users can also select other languages from a list.

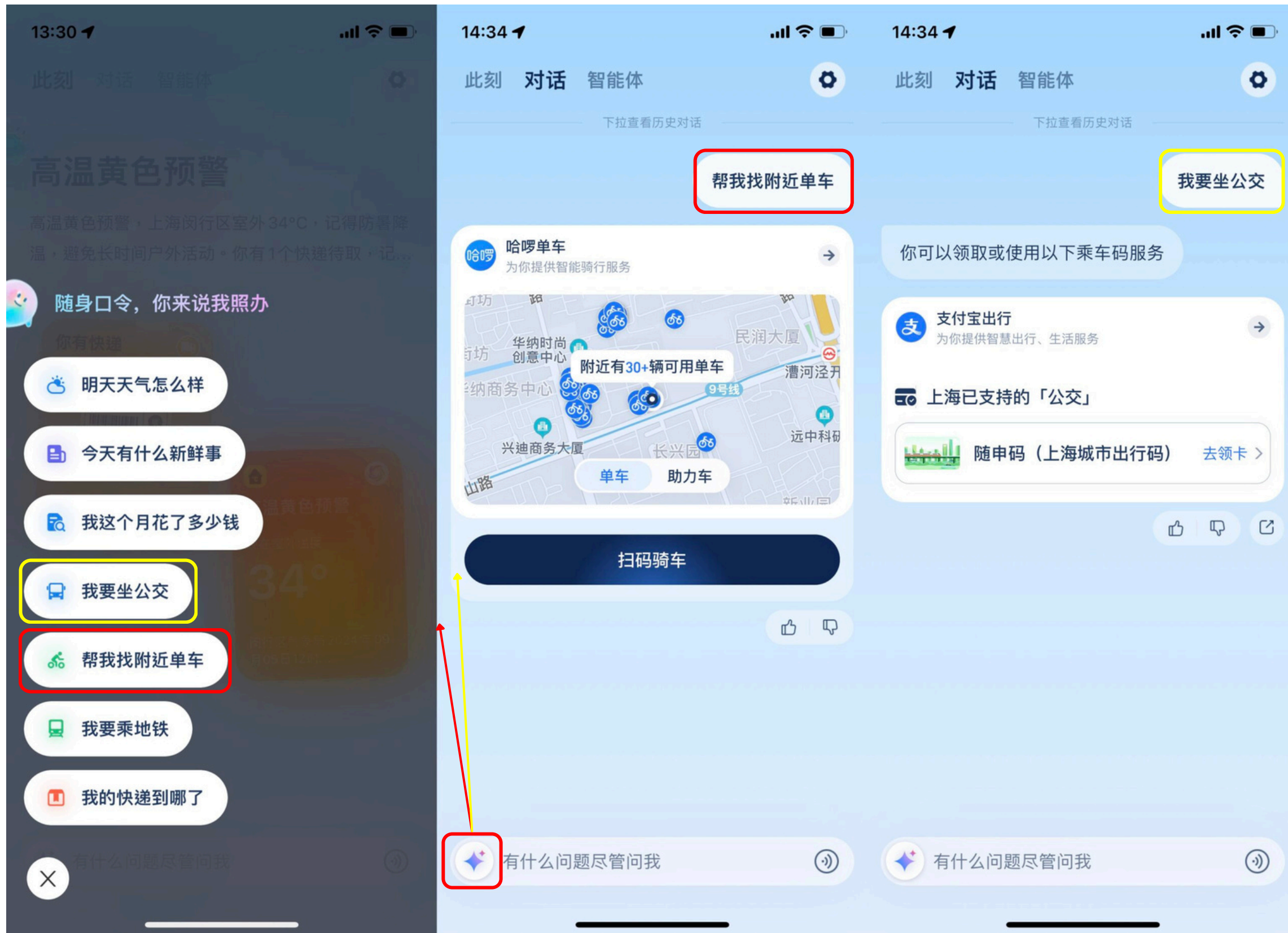
Instead of burying this function in the “Language” menu under “Settings,” Alipay made it a separate, floating element.

This provides **two key benefits:**

users can find and use it instantly, and

the development team avoids the complex task of creating entirely new interfaces for each language.





The left side of the input bar represents “**Quick Commands.**”

It includes common commands such as checking the weather, viewing daily news, reviewing monthly expenses, and finding nearby shared bikes.

When we select “**Find a nearby bike,**” the service provider is **Hello Bike.**

Within the **chat window**, users can scan a **QR code** to start riding directly.

If users tap the arrow on the right side of the service card, they are redirected to the corresponding Alipay mini-program for further actions.

Since we're currently in Singapore, and this app can only locate bikes in mainland China, we don't see any bike information on our page, so we haven't included a screenshot from our own phone.



From the [website](#)

Zhixiaobao lets **users complete common tasks through an AI assistant using cards and quick commands, instead of searching through menus.**

It also demonstrates **a good balance between fast actions in chat and full control through service pages**, which is a key design challenge in large multi-service apps.

Key Insights:

Zhixiaobao shows **how an AI assistant can act as a unified entry point for a super-app with many services.**

By combining chat, cards, and smart actions, it reduces the need for users to search through complex menus.

AI is not presented as a separate feature, but as a new way to access and operate the super-app.

Zhixiaobao surfaces AI capabilities through a conversation-based assistant embedded inside Alipay.

Users express their intent in natural language, and the system responds with actionable cards, buttons, and shortcuts instead of long text answers.

Each card links directly to an existing Alipay service (such as payments, transport, or food delivery), turning the assistant into a service navigation and execution layer rather than a standalone chatbot.

When we abstract AI use and roles

-across different domains -

payments, maps, feeds, logistics and marketplaces

-and across AI design guidelines -

Microsoft Hax AI guideline, Google PAIR AI guideline

We see a repeated pattern. AI is used to -

(a) coordinate complexity

(b) shape choices

(c) remove/reduce work

- <https://www.microsoft.com/en-us/haxtoolkit/>
- [Pair.withgoogle.com](https://pair.withgoogle.com)
- <https://www.salesforce.com/marketing/personalization/ai/>
- <https://www.salesforce.com/ap/artificial-intelligence/ai-automation/>
- <https://zapier.com/blog/ai-orchestration/>
- <https://martech.org/how-visible-and-invisible-ai-shape-the-customer-experience/>



When we abstract AI use and roles

-across different domains -

payments, maps, feeds, logistics and marketplaces

-and across AI design guidelines -

Microsoft Hax AI guideline, Google PAIR AI guideline

We see a repeated pattern. AI is used to -

(a) coordinate complexity

(b) shape choices

(c) remove/reduce work

Accordingly, there are 3 **ROLES** which AI can take up :

- **Orchestration (co-ordinate)**
- **Personalization (suggest)**
- **Automation (act)**

- <https://www.microsoft.com/en-us/haxtoolkit/>
- [Pair.withgoogle.com](https://pair.withgoogle.com)
- <https://www.salesforce.com/marketing/personalization/ai/>
- <https://www.salesforce.com/ap/artificial-intelligence/ai-automation/>
- <https://zapier.com/blog/ai-orchestration/>
- <https://martech.org/how-visible-and-invisible-ai-shape-the-customer-experience/>



Role	Core function	Typical visibility	Responsibility
Orchestration	Coordinates people, time, resources, matching, scheduling, load balancing, forecasting demand	Invisible, visible only when broken	Makes the system run
Personalization	Shapes what each user sees & gets, discovery, ranking, recommendation, decision support	Semi-visible, visible enough to feel helpful	Shapes choices without deciding
Automation	Acts, filters, blocks, executes, generates, detects fraud, flags behaviour	Semi-visible, visible enough to feel helpful	Acts on user's behalf (or stops user)

Google Maps

UX goal: **Don't interrupt the journey, improve it**

Role 1 - AI as **Orchestrator**, Visibility: **Invisible**

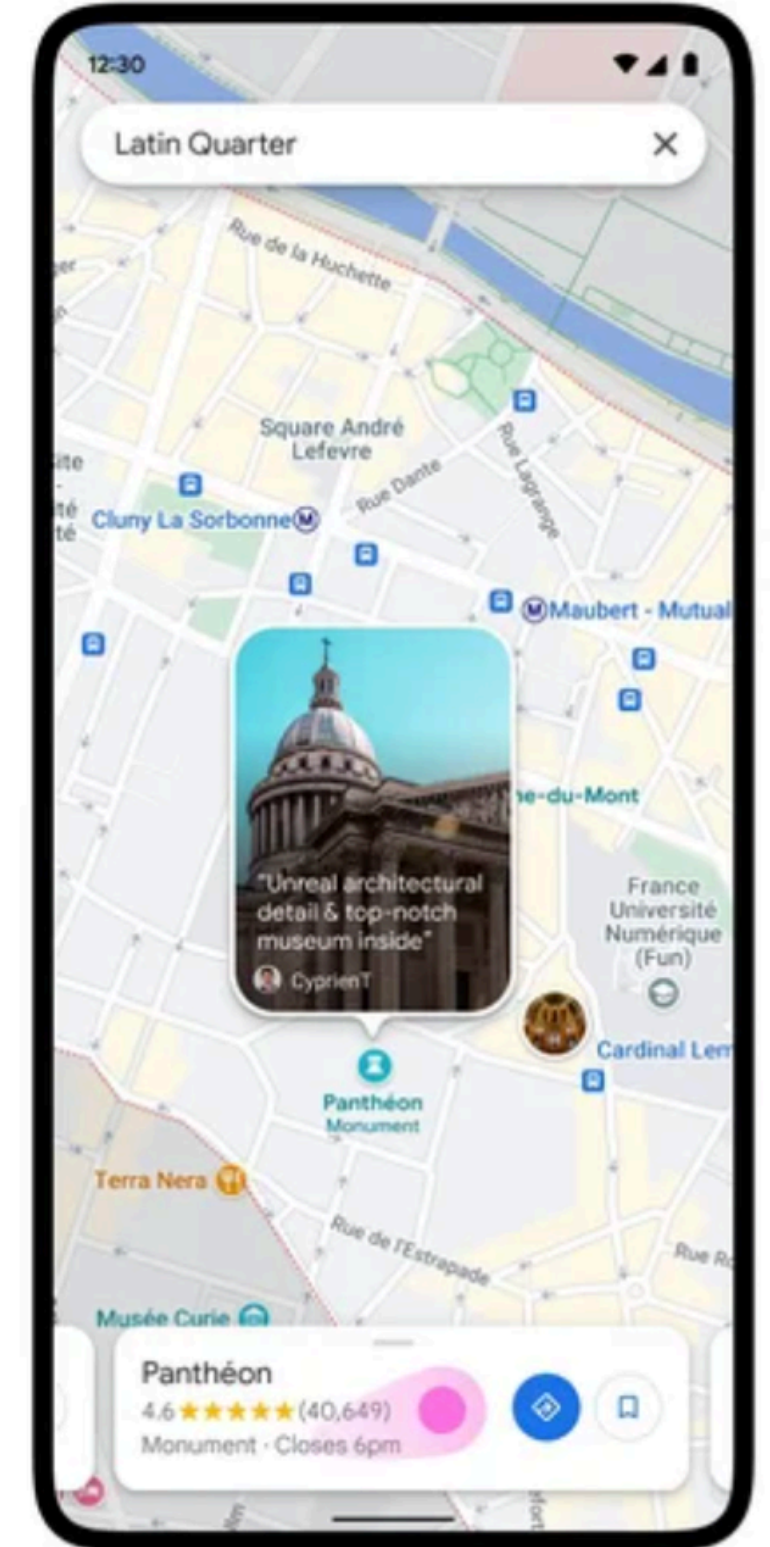
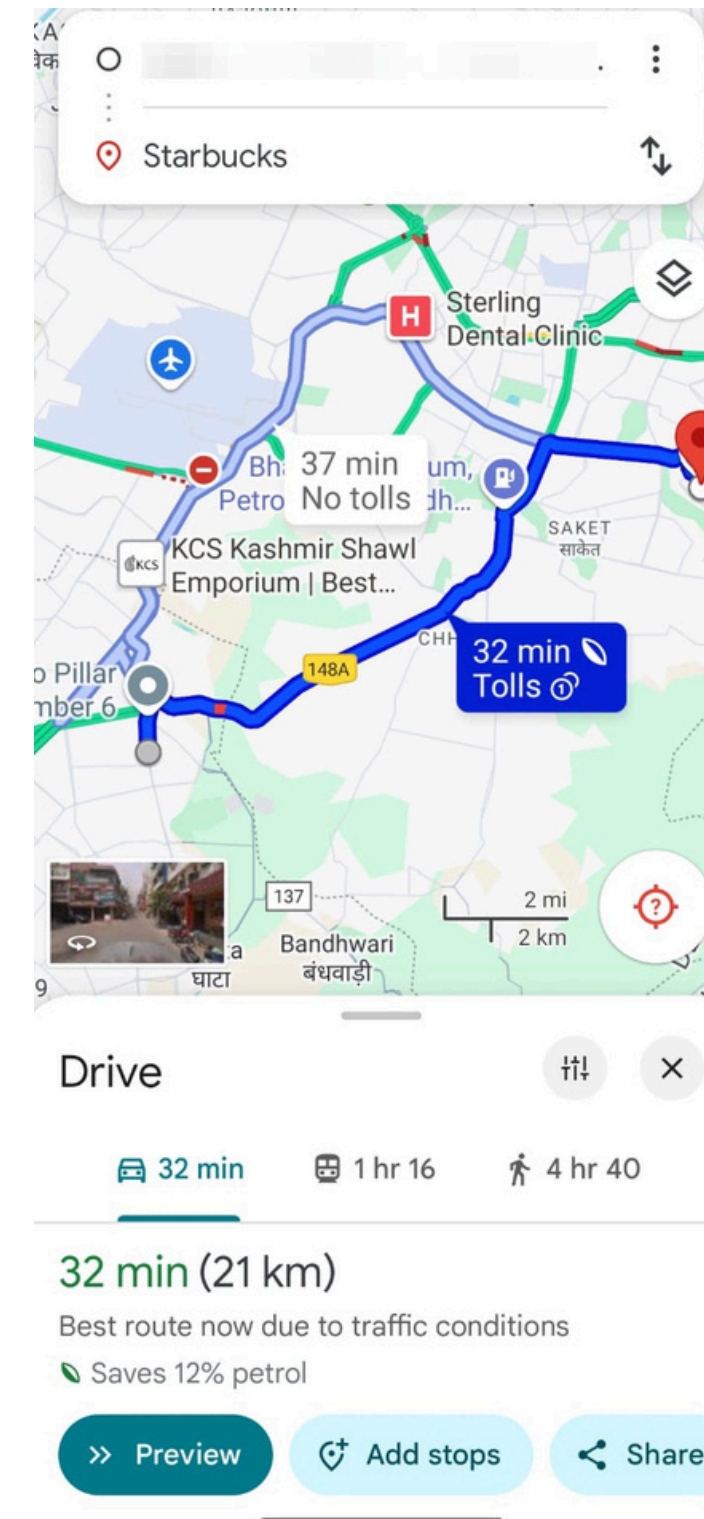
- Coordinates millions of drivers simultaneously
- AI predicts traffic, reroutes for jams
- AI analyzes vast amounts of imagery to detect changes, identify features and keep the map fresh.

Role 2 - AI as **Personalization**, Visibility: **Semi visible**

- Mode preferences (walking vs driving vs public transport)
- Place recommendations (restaurants, cafes, saved spots)

Role 3 - AI as **Automation**, Visibility: **Semi visible**

- Automatic rerouting when traffic changes
- Speed trap / incident detection
- ETA updates without user input



<https://www.analyticsvidhya.com/blog/2024/03/google-maps-ai-features-that-you-must-know/>

Uber

UX goal: **Users experience *shorter waits*.**

Role 1 - AI as **Orchestrator**, Visibility: **Invisible**

- Driver-rider matching across segments
- Demand prediction across segments
- City-scale optimization and dynamic pricing
- Graceful degradation - Clear status when coordination breaks

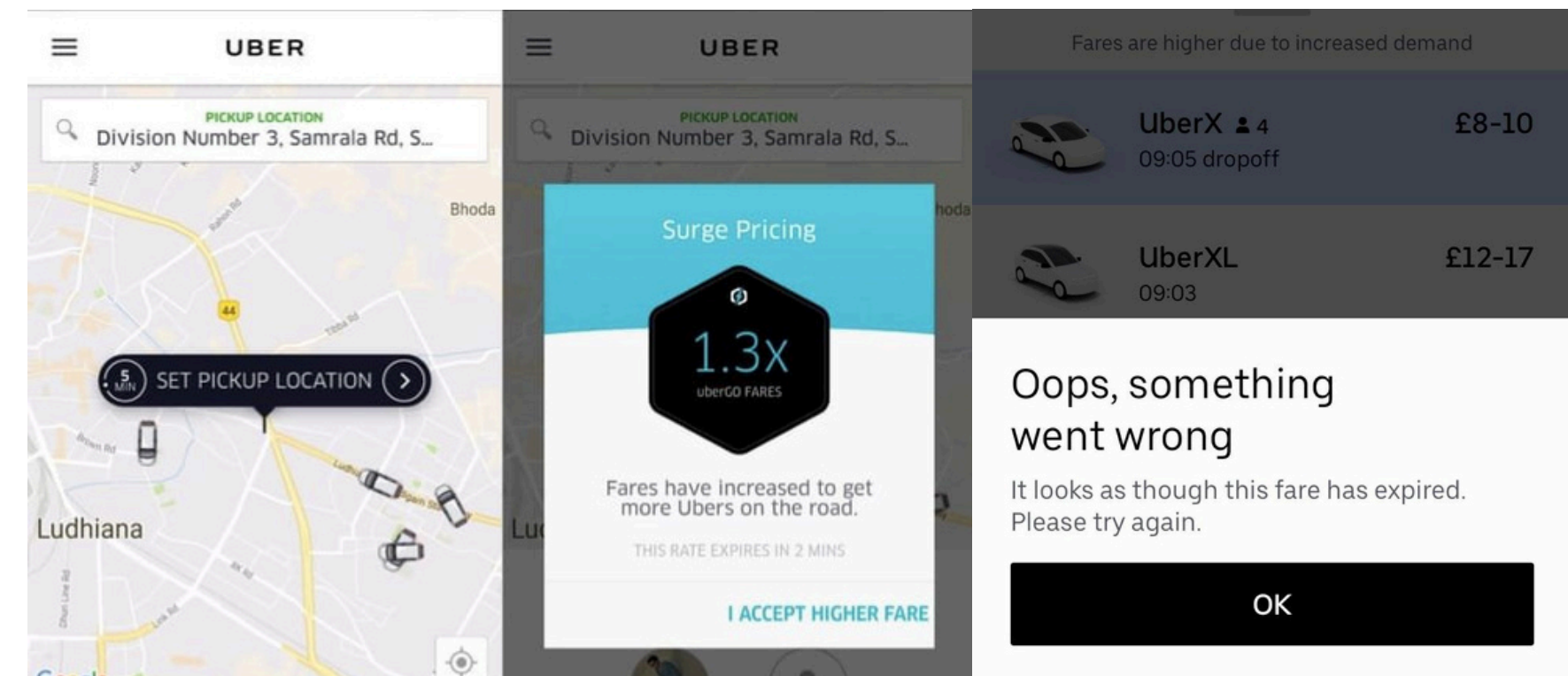
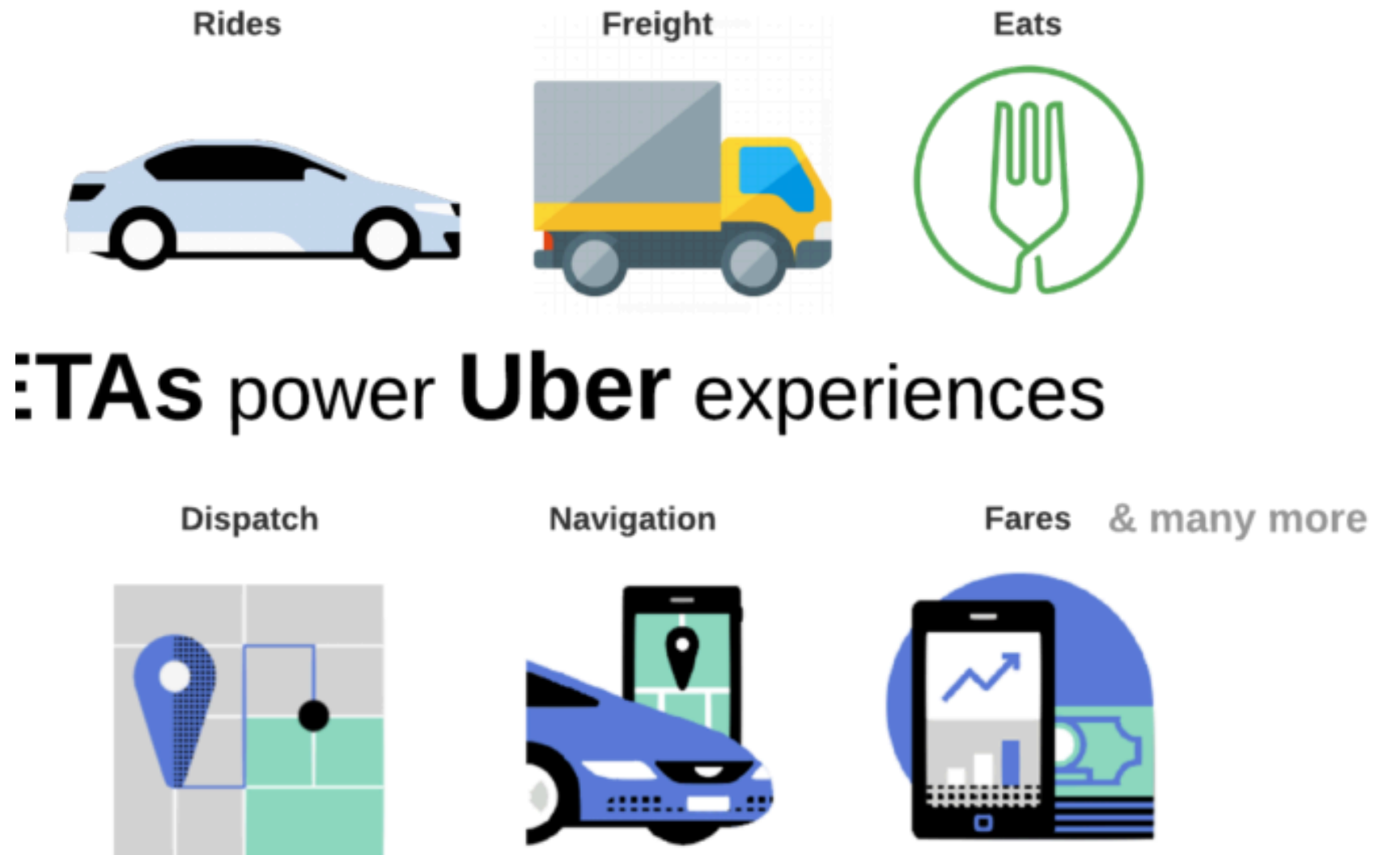
Role 2 - AI as **Personalization**, Visibility: **Semi visible**

- Suggested pickup points
- Ride type ranking (UberX, Premier, Auto, Pool)
- Habit learning (frequent routes, commute timing)

Role 3 - AI as **Automation**, Visibility: **Semi visible**

- Live ETA recalculation, automatic rerouting during traffic
- Trip progress monitoring
- Safety anomaly detection (long stops, route deviation)

<https://www.uber.com/blog/engineering/ai/>



Netflix

UX goal: **Reducing decision fatigue.**

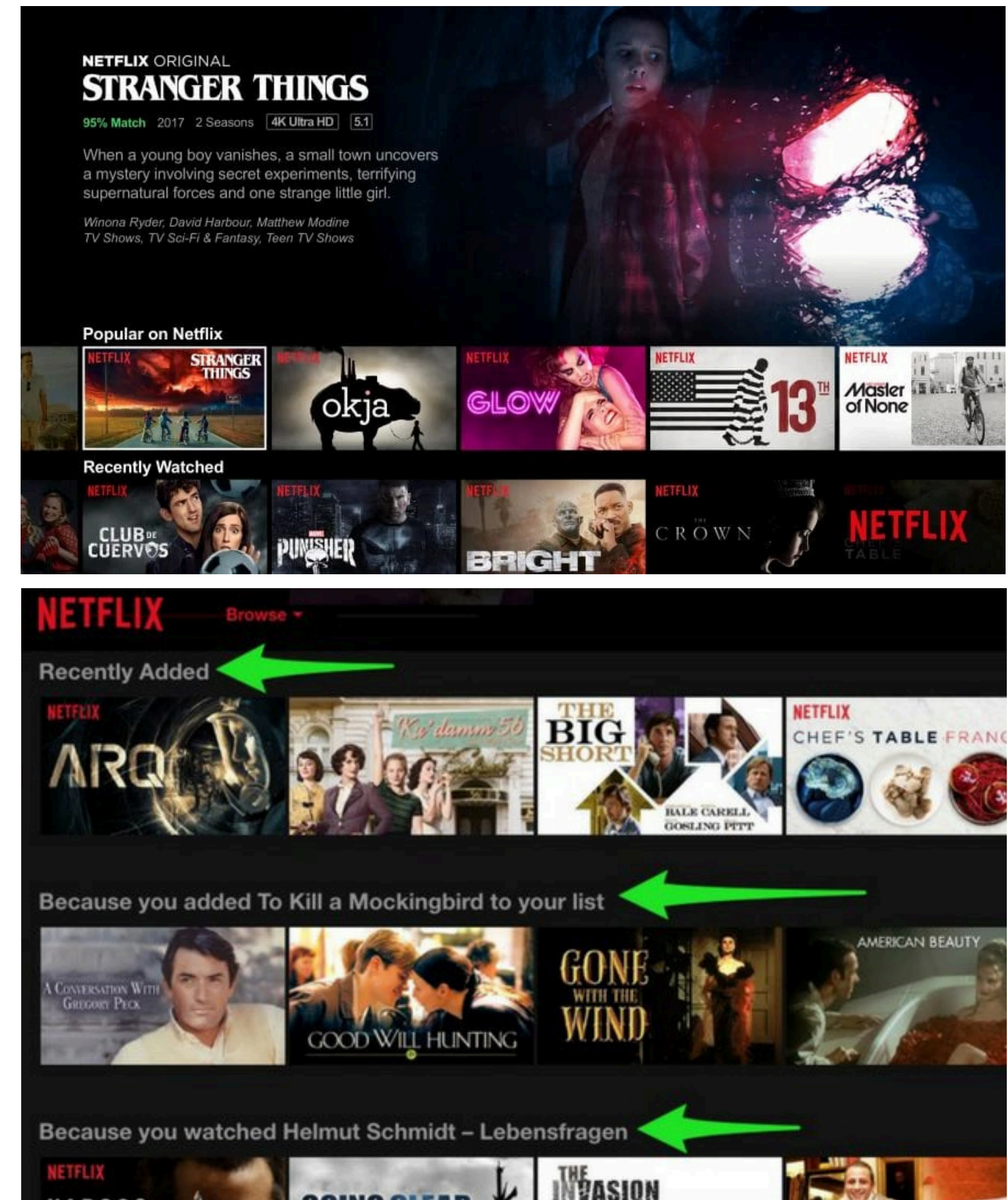
Role 2 - AI as **Personalization**, Visibility: **Semi visible**

- Over 80% of content watched on Netflix is driven by its AI-powered recommendation engine, which analyzes user viewing habits, pauses, skips, and search history.
- AI selects different thumbnails and artwork for the same show to match the preferences of individual users
- Used for fast, high-quality subtitling and voiceover dubbing in multiple languages.

Role 3 - AI as **Automation**, Visibility: **Semi visible**

- Autoplay next episode
- Skip intro / recap
- Resume across devices
- Dynamic buffering & quality adjustment

<https://digitaldefynd.com/IQ/ways-netflix-uses-ai/>



Spotify

UX goal: **Hyper personalize song suggestions.**

Role 1 - AI as **Orchestrator**, Visibility: **Invisible**

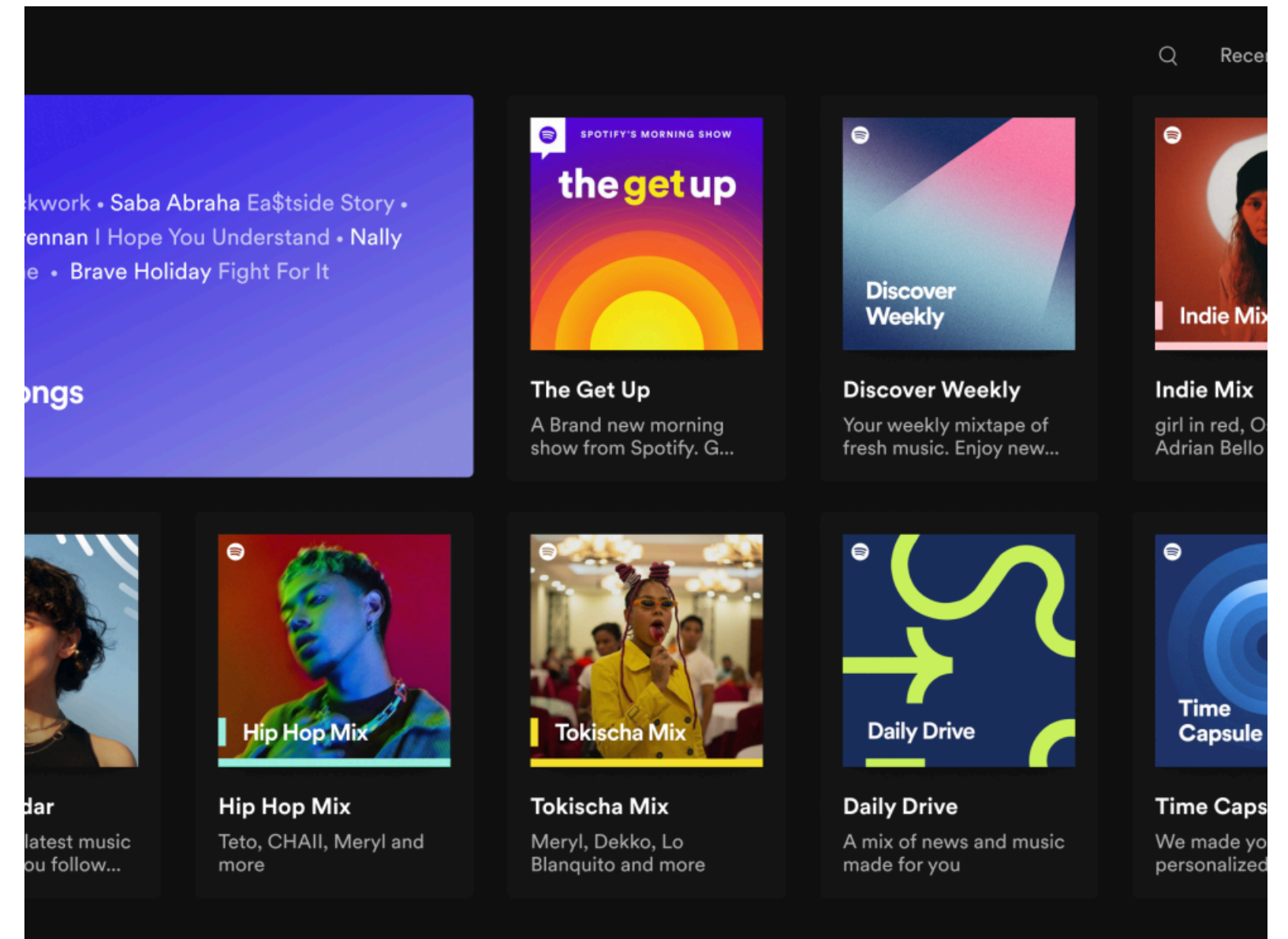
- Balancing artist exposure and platform constraints
- Traffic & usage forecasting

Role 2 - AI as **Personalization**, Visibility: **Semi visible**

- Taste profiling
- Context inference (time, activity, mood)
- Personalized mixes, "Daylist" playlists, and song suggestions
- Uses OpenAI tech for song discovery

Role 3 - AI as **Automation**, Visibility: **Semi visible**

- Autoplay after a playlist ends
- Seamless transitions between tracks
- Cross-device session continuation
- Smart shuffle ordering



- <https://www.marketingainstitute.com/blog/spotify-artificial-intelligence>

Amazon

UX goal: Improve efficiency, and enhance customer experience.

Role 1 - AI as **Orchestrator**, Visibility: **Invisible**

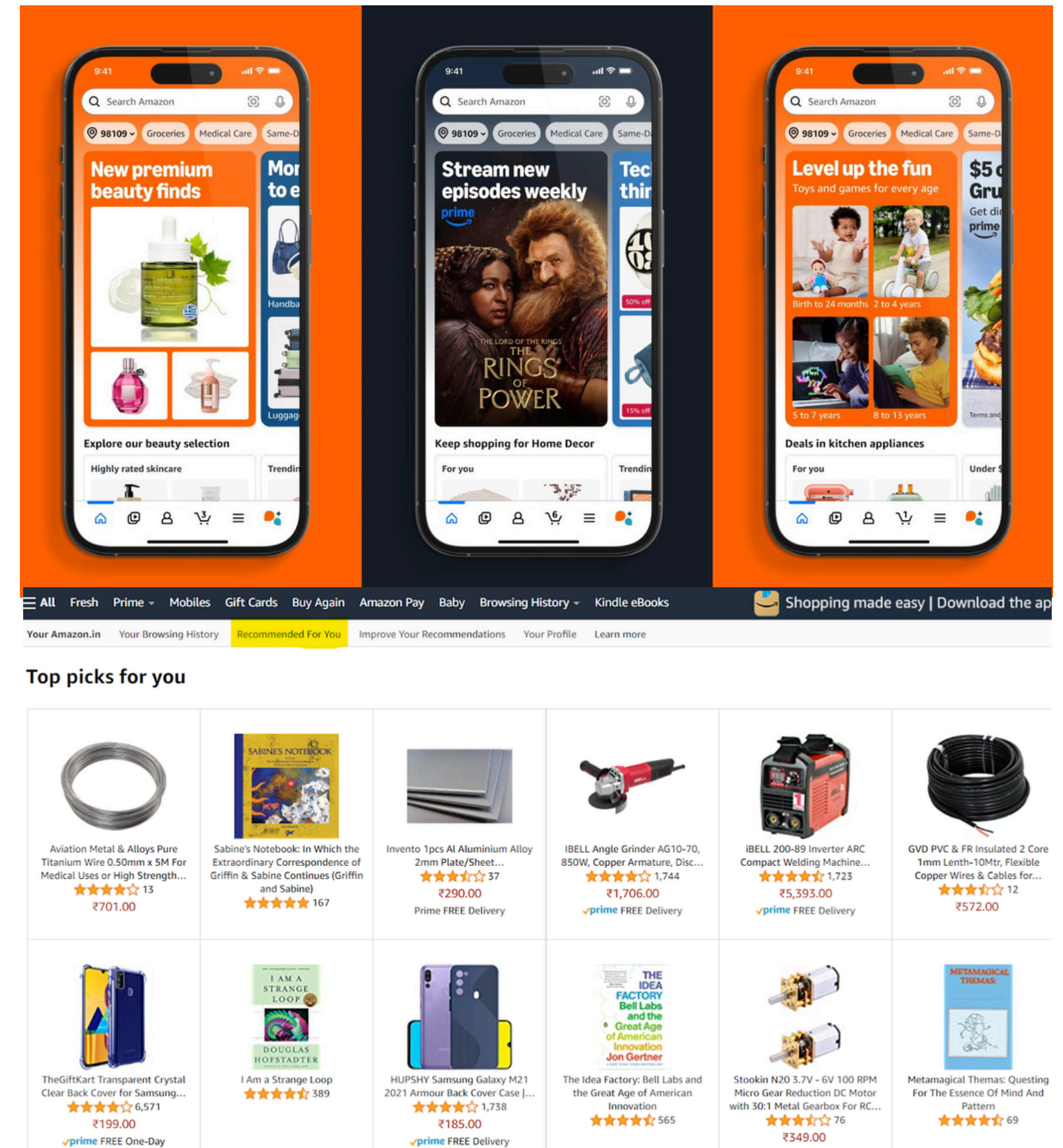
- Predicting demand before orders happen
- Deciding which warehouse should fulfill your order
- Coordinating robots, pickers, trucks, last-mile delivery
- Load balancing during sales (Prime Day)

Role 2 - AI as **Personalization**, Visibility: **Semi visible**

- Cross-sell nudging
- Analyze browsing history to provide personalized product recommendations
- Price sensitivity modeling

Role 3 - AI as **Automation**, Visibility: **Semi visible**

- One-click checkout
- Automatic refunds & returns
- Fraud detection



- <https://www.aboutamazon.com/news/innovation-at-amazon/amazon-generative-ai-seller-growth-shopping-experience>

Gmail

UX goal: **Increase productivity**

Role 1 - AI as **Orchestrator**, Visibility: **Invisible**

- Routing billions of emails reliably
- Managing load spikes (news, sales, spam waves)
- Synchronizing messages across devices

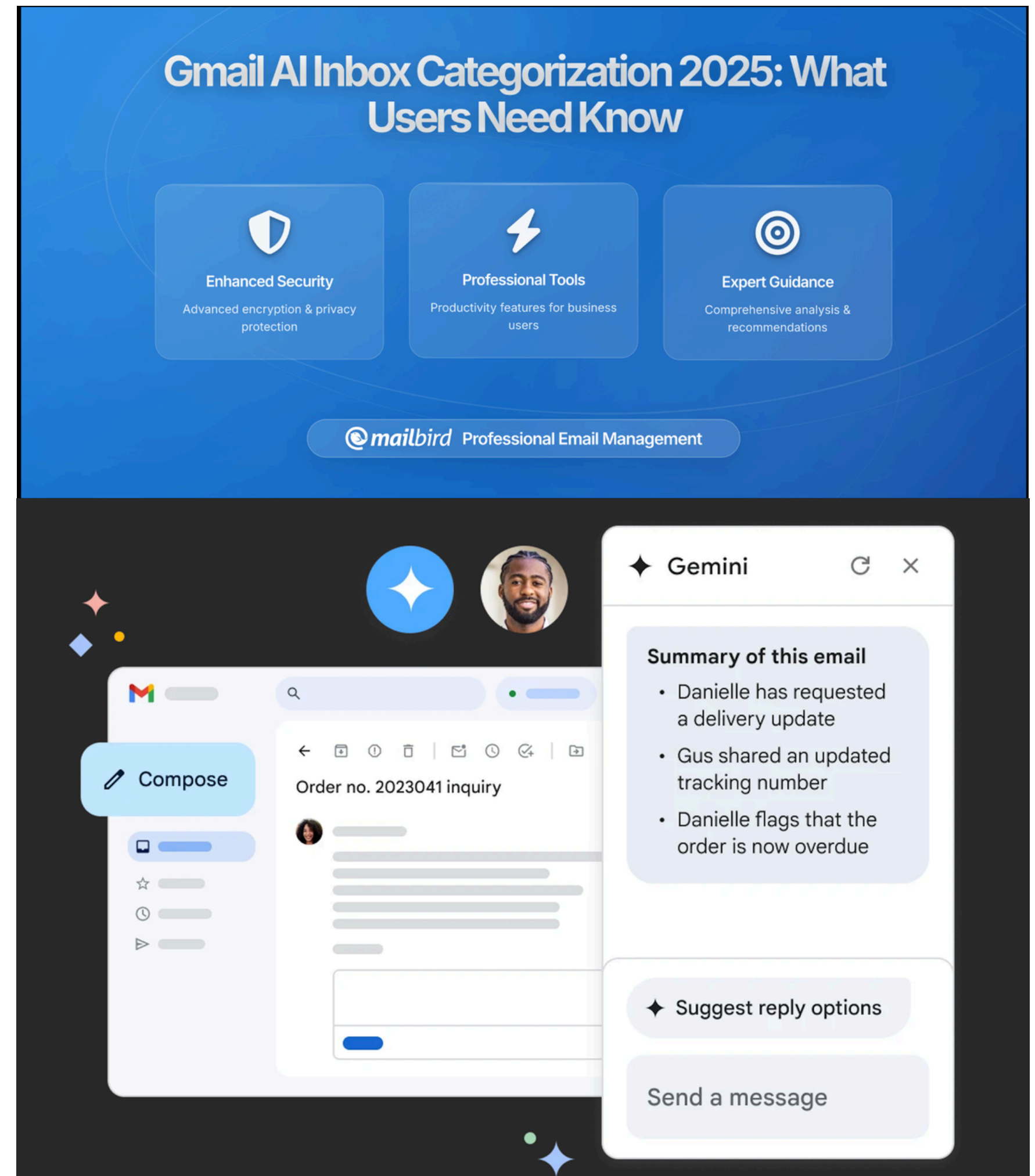
Role 2 - AI as **Personalization**, Visibility: **Semi visible**

- Priority Inbox (important vs everything else)
- Highlighting important senders, behaviour-based filtering
- Nudging reminders (“Reply?” / “Follow up?”)

Role 3 - AI as **Automation**, Visibility: **Semi visible**

- Summarizes emails to help users catch up on key points
- Smart Reply suggests detailed, personalized responses
- Block spam, phishing, and malware, filtering out unwanted messages

- https://workspace.google.com/intl/en_my/products/gmail/ai/#:~:text=Write%20emails%20faster%20with%20smart,keeping%20your%20inbox%20under%20control.



Visa

UX goal: **Secure, optimize, and modernize the global payment ecosystem**

Role 1 - AI as **Orchestrator**, Visibility: **Invisible**

- Coordinating transactions across banks, countries, currencies
- Managing peak loads (sales, festivals, global events)
- Ensuring near-zero downtime worldwide

Role 2 - AI as **Personalization**, Visibility: **Invisible**

- Learning spending patterns per card
- Understanding location, merchant type, frequency
- Adjusting risk tolerance dynamically

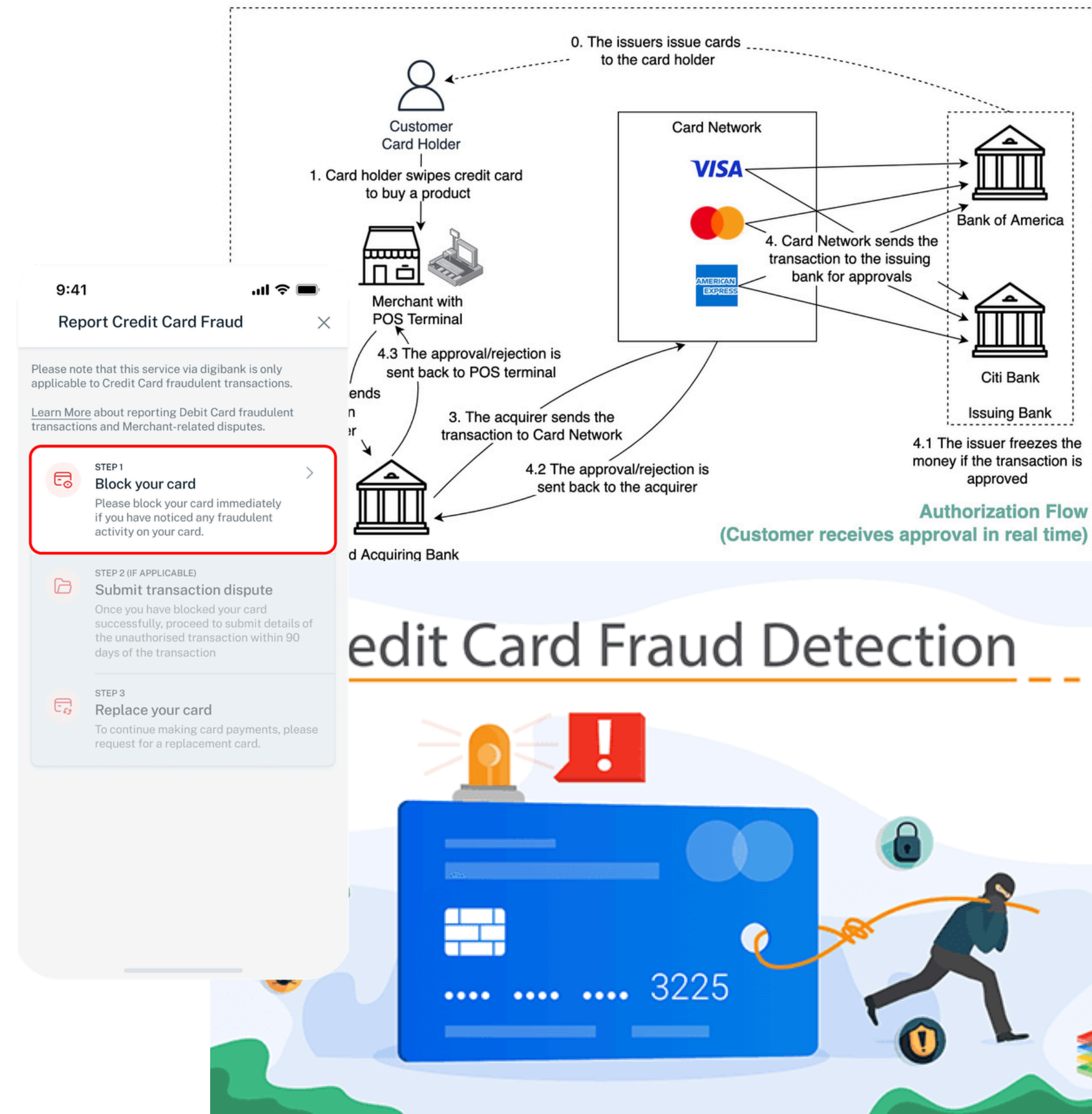
Role 3 - AI as **Automation**, Visibility: **Semi visible**

- Evaluates fraud risk in real-time for transaction
- Spot bot-driven attacks
- Escalation triggering
- Clear paths for recovery when AI fails

- <https://corporate.visa.com/en/sites/visa-perspectives/trends-insights/ai-the-next-frontier.html#:~:text=All%20in%20when%20it%20comes,advance%20its%20gen%20AI%20initiatives.>

How does VISA Work?

blog.bytebytego.com



Superapp – Grab (Singapore)

UX goal: a seamless, user-centric "everyday superapp". Began as ride-hailing and expanded into food delivery, courier services, and financial services

Role 1 - AI as **Orchestrator**, Visibility: **Invisible**

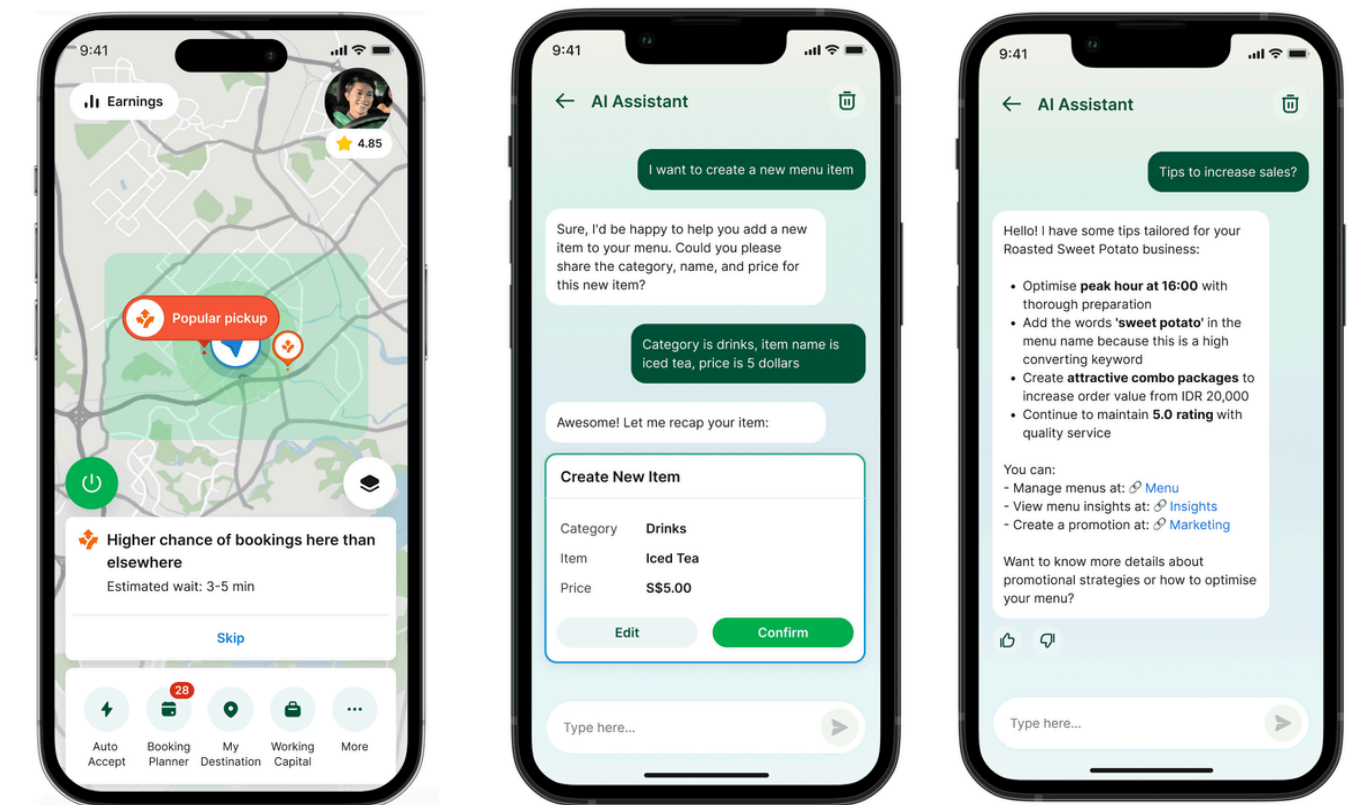
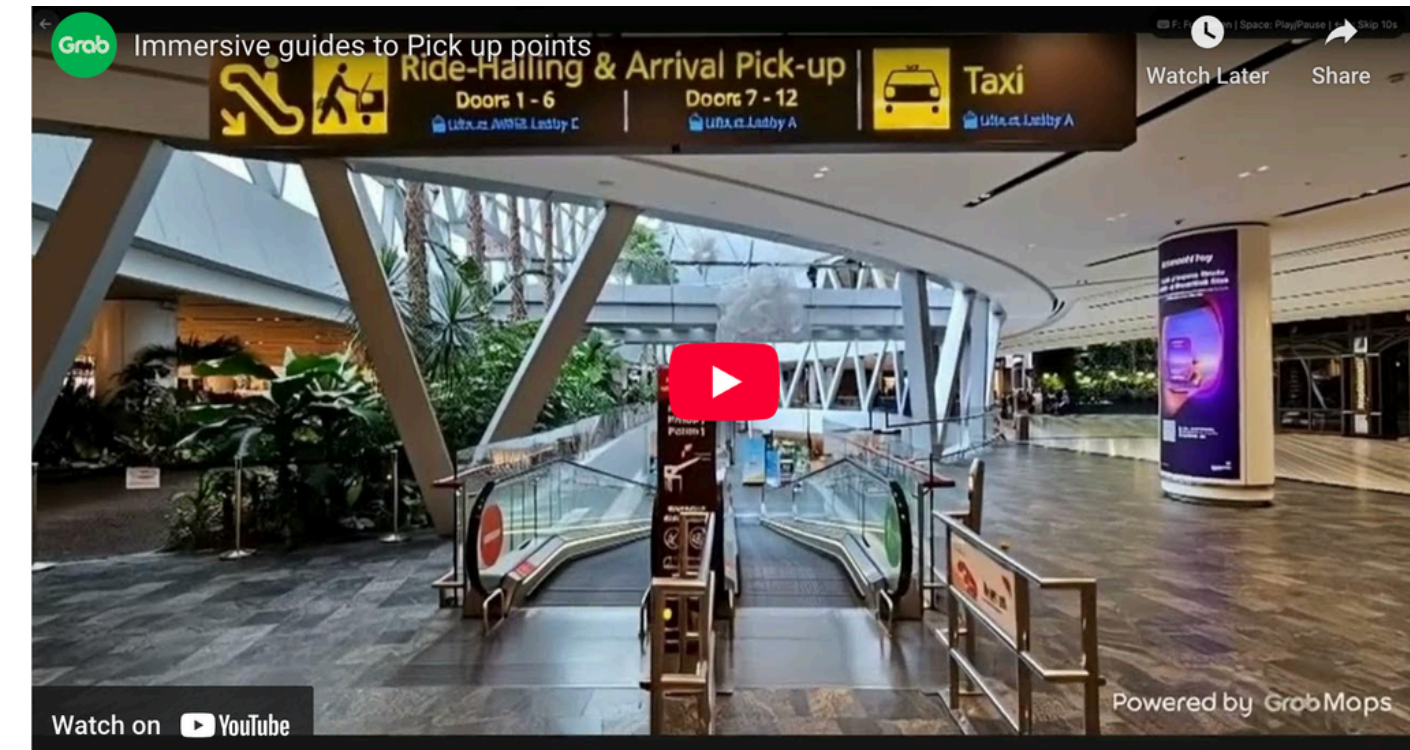
- Ensuring delivery reliability for large orders with automated order splitting, This is done by a system-initiated split where a Large Language Model evaluates orders based on size and weight
- AI-assisted voice reporting tool enables drivers to update Grab about road conditions in real time.

Role 2 - AI as **Personalization**, Visibility: **Invisible**

- Overcoming language barriers with accurate local menu translations
- AI Voice Assistant allows visually impaired users to simply speak their intent and get going
- The AI Merchant Assistant seeks to level the playing field by serving as every merchant's 24/7 business advisor.

Role 3 - AI as **Automation**, Visibility: **Semi visible**

- Solving navigation friction with visual pickup guides - videos created by AI using static images, can quickly roll out video guides for more locations.
- keeps users updated on the driver's status, from the moment a driver is found to their arrival.
- Grab's new AI ride-guidance feature guides drivers towards hotspots more quickly and precisely to find jobs faster.



- <https://www.grab.com/sg/inside-grab/stories/four-ways-grab-solves-real-world-user-challenges-with-ai/>
- <https://investors.grab.com/news-and-events/news-details/2025/Grab-deploys-agentic-AI-to-empower-merchants-and-driver-partners-2025-KuA890N398/>

Superapp – We Chat (China)

UX goal: **friendly, efficient, and cohesive digital ecosystem**. Began as Messaging and expanded into Social Media, Payments, Mini Programs, work

Role 1 - AI as **Orchestrator**, Visibility: **Invisible**

- seamlessly connecting conversations with content discovery and social sharing.
This integration is reducing friction in user journeys
- AI can help in group chats, like making things more efficient and adding a sense of companionship.

Role 2 - AI as **Personalization**, Visibility: **Invisible**

- Integrate chatbots to handle customer inquiries, recommend products, and track orders 24/7.
- News and video mini programs use algorithms to analyze user behavior and serve personalized content

Role 3 - AI as **Automation**, Visibility: **Semi visible**

- AI agents handle initial, routine customer inquiries, with capabilities to escalate complex issues to human agents.
- Share access to customer histories, and assign conversations to the right colleagues.
- A tool for generating images based on text prompts.



- <https://aioproem.substack.com/p/ai-enhancing-wechats-super-app-status>
- <https://www.omnichat.ai/sg/wechat/#:~:text=Automation,time%20and%20reducing%20service%20workload.>
- <https://valuechina.net/en/china-blog/social-media-en/wechat-the-ultimate-guide-for-marketing/#:~:text=WeChat%20Search%20has%20been%20enhanced%20with%20AI,for%20generating%20images%20based%20on%20text%20prompts.>
- <https://technode.com/2026/01/27/ai-gets-social-in-china-tencent-tests-yuanbao-groups-for-ai-powered-social-interaction/#:~:text=Details:%20Tencent%20is%20taking%20AI,link%20or%20a%20group%20ID.>

Superapp – Paytm (India)

UX goal: a seamless, user-centric "everyday superapp". Began as Mobile Payments and expanded into E-commerce, Bill Payments, Ticketing and Travel

Role 1 - AI as **Orchestrator**, Visibility: **Invisible**

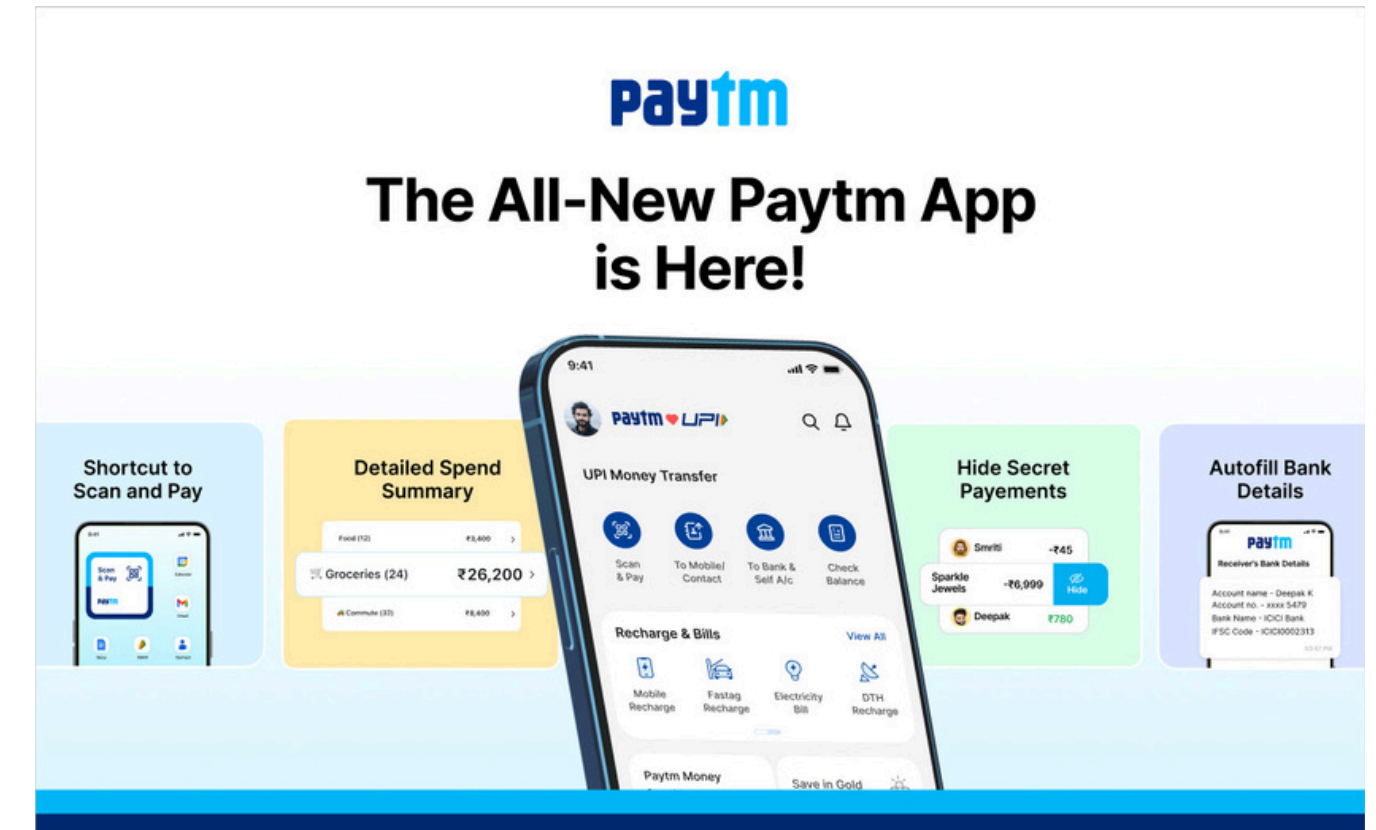
- AI-powered POS devices provide real-time, voice-based updates on transactions and merchant performance insights.
- Paytm ARMS (merchant lifecycle insights platform) and Paytm Pi (fraud and risk detection system) automate merchant onboarding, fraud detection, segmentation, and pricing optimization.

Role 2 - AI as **Personalization**, Visibility: **Invisible**

- AI brings fun to finance with Paytm Playback, a unique feature that turns your recent transactions into a personalized rap song, making money management engaging
- The company uses conversational AI agents to provide personalized assistance to customers via text and voice across 11 languages.

Role 3 - AI as **Automation**, Visibility: **Semi visible**

- Translate natural language into complex database queries
- AI-powered categorization, the Monthly Spend Summary feature organizes your expenses across shopping, bills, travel, and more, simplifying budgeting



- <https://paytm.com/blog/artificial-intelligence/paytms-ai-system-recognised-globally-on-spider-2-0-becomes-first-indian-company-on-the-global-capability-leaderboard/>
- <https://paytm.com/blog/artificial-intelligence/paytm-all-new-app-experience-is-live-here-are-10-features-you-shouldnt-miss/#:~:text=Shouldn't%20Miss-,Paytm%20All%2DNew%20App%20Experience%20Is%20Live%2C%20Here%20Are%2010,Paytm%20UPI%2Dlinked%20bank%20accounts.>
- <https://economictimes.indiatimes.com/tech/technology/how-paytm-is-using-ai-for-in-house-ops/articleshow/122863533.cms?from=mdr>